

Cloud Print Server

PowerShell is the practical way to handle this on a Windows domain network.

Google Cloud Print was discontinued in 2020. Modern equivalents are **Microsoft Universal Print** (cloud-native) or a traditional on-premises/hybrid print server that can be fronted by cloud services. Below are production-ready approaches.

1. Recommended: Client-side mapping script (works with any print server or Universal Print shares)

Deploy this via Group Policy (User Configuration → Logon Script), Intune, or a scheduled task that runs at user logon. It maps printers based on AD group membership and installs the connection (drivers are pulled from the print server via Point and Print if configured correctly).

```
# Auto-map network / cloud printers for domain users
# Run as the logged-on user (not SYSTEM)

# ===== CONFIGURATION =====
$PrintServer = "printserver.contoso.com" # or IP / Universal Print share
name
$LogPath      = "$env:TEMP\PrinterMapping.log"

# Define printers: ShareName, AD Group (empty = all users), Default?
$Printers = @(
    @{ Share = "Reception";   Group = "";                      Default = $true
    },
    @{ Share = "Accounting";  Group = "GG-Accounting";        Default = $false
    },
    @{ Share = "Warehouse";   Group = "GG-Warehouse";         Default = $false
    },
    @{ Share = "HR-Secure";    Group = "GG-HR";                Default = $false
    }
)

# ===== FUNCTIONS =====
function Write-Log {
    param([string]$Message)
    $timestamp = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    "$timestamp - $Message" | Out-File -FilePath $LogPath -Append -Encoding
    UTF8
    Write-Host $Message
}

function Test-ADGroupMembership {
    param([string]$GroupName)
    if ([string]::IsNullOrEmpty($GroupName)) { return $true }
    try {
        $groups = whoami /groups /fo csv | ConvertFrom-Csv
    }
```

```

        return $groups.'Group Name' -contains $GroupName -or
               $groups.'Group Name' -like "*\$GroupName"
    } catch {
        Write-Log "WARNING: Could not check group membership for $GroupName"
        return $false
    }
}

# ===== MAIN =====
Write-Log "Starting printer mapping for $env:USERNAME"

foreach ($p in $Printers) {
    $connectionName = "\\$PrintServer\$($p.Share)"

    if (-not (Test-ADGroupMembership -GroupName $p.Group)) {
        Write-Log "Skipping $($p.Share) - user not in group $($p.Group)"
        continue
    }

    # Check if already mapped
    $existing = Get-Printer -Name $connectionName -ErrorAction
SilentlyContinue
    if ($existing) {
        Write-Log "Already mapped: $connectionName"
    } else {
        try {
            Add-Printer -ConnectionName $connectionName -ErrorAction Stop
            Write-Log "Successfully mapped: $connectionName"
        } catch {
            Write-Log "ERROR mapping $connectionName : $
($_.Exception.Message)"
            # Fallback for older systems
            try {
                (New-Object -ComObject
WScript.Network).AddWindowsPrinterConnection($connectionName)
                Write-Log "Mapped via COM fallback: $connectionName"
            } catch {
                Write-Log "COM fallback also failed for $connectionName"
            }
        }
    }
}

# Set as default if requested

```

```

if ($p.Default) {
    try {
        $printer = Get-Printer -Name $connectionName -ErrorAction SilentlyContinue
        if ($printer) {
            (Get-WmiObject -Class Win32_Printer -Filter "Name='$($connectionName.Replace('\','\\'))'").SetDefaultPrinter()
            # Or modern way:
            # Set-Printer -Name $connectionName -IsDefault $true # requires newer PowerShell
            Write-Log "Set as default: $connectionName"
        }
    } catch {
        Write-Log "Could not set default printer: $($_.Exception.Message)"
    }
}

Write-Log "Printer mapping completed"

```

Deployment notes:

- For **Point and Print** restrictions (security), configure the GPO:
Computer Configuration → Policies → Administrative Templates → Printers → Point and Print Restrictions
Allow the print server and set “Do not show warning or elevation prompt”.
- Drivers must already exist on the print server (upload via Print Management console).
- Test with a single user first.

2. Microsoft Universal Print (true cloud solution)

If you want a pure cloud approach (no on-prem print server):

1. Install the **Universal Print connector** on a machine that can reach the physical printers.
2. Register printers in the Azure portal / Universal Print admin center.
3. Share them and assign to Entra ID groups or users.
4. On clients (Windows 10/11, Azure AD joined or hybrid), users simply go to
Settings → Bluetooth & devices → Printers & scanners → Add device → Search

Universal Print.

PowerShell management (admin side):

```

Install-Module UniversalPrintManagement -Scope CurrentUser
Connect-UPService

# List printers
Get-UPPrinter

# Grant access to a group
$share = Get-UPPrinterShare | Where-Object { $_.DisplayName -eq "Reception" }
Grant-UPAccess -PrinterShareId $share.Id -GroupId ""

```

Clients can also be targeted via Intune configuration profiles or scripts that call the Universal Print discovery.

3. Server-side / appliance-style approach

If the “printer appliance” is a Windows machine acting as a print server that itself needs to stay connected:

- Use the Universal Print connector (official Microsoft component).
- Or run a small service that keeps the connection alive and publishes printers via IPP or SMB.

For driver pre-staging on the print server (so clients get them automatically):

```
# On the print server (run as admin)
Add-PrinterDriver -Name "HP Universal Printing PCL 6" -InfPath "C:\Drivers\hpcu250u.inf"
# Then assign the driver to the shared printer via Print Management or:
Set-Printer -Name "Reception" -DriverName "HP Universal Printing PCL 6"
```

Best practice summary

Scenario	Recommended Method	Code / Tool
Classic domain + on-prem server	Logon script + GPO	Script above
Hybrid / cloud-first	Microsoft Universal Print	Connector + Intune
Full cloud, no servers	Universal Print + Entra groups	Portal + PowerShell module
Legacy Google Cloud Print	Not supported (shut down 2020)	Migrate

Would you like me to adapt the script for:

- Specific printer models / driver names?
- Intune / Endpoint Manager deployment?
- Removal of old printers?
- macOS / Linux clients as well?

Provide the print server name, share names, and any AD groups you use and I can tailor it further.